The final result is $\Delta Y = 100+100\times4/5+100\times[4/5]^2+100\times[4/5]^3$ or,

$\Delta Y = 100 + 100 \text{ x } 0.8 + 100 \text{ x } (0.8)^2 + 100 \text{ x } (0.8)^3$

$= 100 + 80 + 64 + 51.2...$

$= 500$

that is $100\times1/1-4/5$

$100\times1/1/5$

$100\times5 = ₹500$ crores

For instance if $C = 100 + 0.8Y$, $I = 100$,

Then $Y = 100 + 0.8Y + 100$

$1y - 0.8y = 100 + 100$

$0.2Y = 200$

$Y = 200/0.2 = 1000 \rightarrow$ Point B

If I is increased to 110, then

$0.2Y = 210$

$Y = 210/0.2 = 1050 \rightarrow$ Point D

For ₹10 increase in I, Y has increased by ₹50.

This is due to multiplier effect.

At point A, $Y = C = 500$

i.e Autonomous consumption 100 + MPC 400

$C = 100 + 0.8\ (500) = 100 + 400 = 500$; $S = 0$

At point B, $Y = 1000$

$C = 100 + 0.8\ (1000) = 900$; $S = 100 = I$

At point D, $Y = 1050$

$C = 100 + 0.8\ (1050) = 940$; $S = 110 = I$

When I is increased by 10, Y increases by 50.

This is multiplier effect ($K = 5$)

$K = \frac{1}{0.2} = 5$

4.4.4 Classification of Multiplier:

1. Static and dynamic multiplier

i. **Static multiplier** is otherwise known as simultaneous multiplier, timeless multiplier, and logical multiplier. Under static multiplier the change in investment and the resulting change in income are simultaneous.There is no time lag. There is also no change in MPC as the economy moves from one equilibrium position to another.

ii. **Dynamic multiplier** is also known as 'sequence multiplier'. In real life, income level does not increase instantly with investment. In fact, there is a time lag between increase in income and consumption expenditure.

4.4.5. Leakages of multiplier

The multiplier assumes that those who earn income are likely to spend a proportion of their additional income on consumption. But in practice, people tend to spend their additional income on other items. Such expenses are known as leakages.

Payment towards past debts.

If a portion of the additional income is used for repayment of old loan, the MPC is reduced and as a result the value of multiplier is cut.

Purchase of existing wealth

If income is used in purchase of existing wealth such as land, building and shares money is circulated among people and never enters into the consumption stream. As a result the value of multiplier is affected.

Import of goods and services

Income spent on imports of goods or services flows out of the country and has little chance to return to income stream in the country. Thus imports reduce the value of multiplier.